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Java
Today
A Better
Tomorrow



Code
Ideas
Progress

Learn
Practice
Build

Programming Language



Language

- Any medium to communicate is called a language.



Programming Language

- Any language that is used to communicate with a computer or a machine is called Programming Language. Programming language is the form of set of instructions.



Types of Programming Language

1

Low Level (Binary Language [0,1]).

- A computer can directly understand assembly language.

2

Assembly Language (Mod, Bit and other)

- It is in the form of mnemonics (easily understood words).
- It is directly converted into machine language by an assembler.



James Gosling and Patrick Naughton

Employees of Sun Microsystems, with support from **Bill Joy** — co-founders of Sun Microsystems.



Low Level Language

- Low level language is the first programming language which was used to instruct computer.
- It is in binary form [0,1], so it is called **Binary Language**.
- A computer can directly understand low level language, but it is difficult for a human to understand.

e.g.:

0	→	off
1	→	on



Assembly Language vs. Machine/Low Level Language

Assembly Language

- A computer can directly understand assembly language.
- It is in the form of mnemonics (easily understood words).
- It is directly converted into machine language by an assembler.

Machine/Low Level Language

- A computer can directly understand machine language.
- It is in the form of 0s and 1s (binary).
- It is not easy for a human to understand.
- It is directly executed by the computer.



Key Point:

A computer cannot understand assembly language directly. So, assembly language is used as a translator which takes assembly language and converts it into low level language (machine language).



High Level Language

- A language that a human can understand directly is called high level language.
- It is a high level language because it is easy to understand.
- It makes programming very easy to understand.

e.g.: 1972 → C language

A computer cannot understand high level language code directly, so compiler and interpreter are used. In C, a translator which takes high level code and converts into low level code is used.



Every high level language will have own compiler.